

Preprocessing Pipeline

CA Assignment 2 · Clustering Algorithms

Embedding model: `sentence-transformers/all-mpnet-base-v2`

Overview

The preprocessing pipeline transforms raw natural language sentences into a compact, geometrically well-structured numerical representation suitable for Agglomerative Hierarchical Clustering (AHC). The pipeline consists of five sequential stages:

1. Data loading and text cleaning
2. Sentence embedding using a pre-trained transformer model
3. L2 normalisation of the raw embedding vectors
4. Adaptive dimensionality reduction via Principal Component Analysis (PCA)
5. Non-linear manifold projection via UMAP

All 1,760 rows are processed, including duplicate sentences. This is a deliberate design decision explained in full below.

1. Data Loading and Text Cleaning

The dataset is stored in a tab-separated file with two columns: an integer identifier (**ID**) and a raw sentence string (**Sentence**). The file is read line by line using a custom parser that identifies record boundaries by detecting lines whose first field is a positive integer followed by a tab character. Any line that does not match this pattern is treated as a continuation of the previous sentence. This approach correctly handles sentences that span multiple physical lines, ensuring all 1,760 instances are loaded.

1.1 Cleaning Strategy

A deliberate **light-touch cleaning strategy** is adopted:

- Removal of escaped quotation marks (`\"` and `\'`).
- Collapsing of repeated whitespace characters into a single space.
- Collapsing of actual newline characters into a single space.
- Stripping of leading and trailing whitespace.
- Replacement of empty strings with the placeholder `'empty'`.

Heavy-handed normalisation (lowercasing, punctuation removal, stopword removal) is intentionally excluded. `all-mpnet-base-v2` is trained on natural, cased, punctuated English text. Its self-attention mechanism is sensitive to capitalisation (e.g., character prefixes such as CORIOLANUS: carry strong identity signals), punctuation, and function words. Removing these would push sentences out of the model's training distribution, degrading embedding quality.

1.2 Duplicate Retention

The dataset contains 64 duplicate rows: 18 unique Notre Dame / Wikipedia sentences each repeated 4–5 times, giving 80 rows total. These duplicates are **intentionally retained** rather than removed.

The reason is geometric: duplicate sentences produce identical embedding vectors, which map to the same point in UMAP space. With `n_neighbors=10`, UMAP constructs a k -nearest-neighbour graph in which each Notre Dame sentence finds most or all of its 10 nearest neighbours within the Notre Dame group itself. This creates a densely connected sub-graph that UMAP projects into a tight, isolated manifold region, which AHC then detects as a distinct cluster.

Without the duplicate rows, the 18 unique Notre Dame sentences represent only 1.1% of the dataset. At that density, each sentence’s 10-nearest-neighbour graph would be dominated by non-Notre-Dame neighbours, the sub-graph would not be cohesive, and the Notre Dame region would dissolve into the surrounding modern English manifold.

Influence on clustering: Retaining duplicates is the mechanism that enables the detection of the Notre Dame sub-domain as a geometrically distinct cluster. Removing them would merge all Notre Dame content into the broader modern English cluster, reducing the clustering to $K = 2$.

2. Sentence Embedding with all-mpnet-base-v2

Sentences are encoded into dense numerical vectors using the pre-trained model `sentence-transformers/all-mpnet-base-v2`, a 12-layer, **768-dimensional** sentence transformer based on the MPNet architecture (Song et al., 2020). The model was fine-tuned on over one billion sentence pairs using a Multiple Negative Ranking (MNR) loss, which optimises the embedding space so that semantically similar sentences map to geometrically proximate points on the unit hypersphere in $\mathbb{R}^{768}$.

2.1 Why This Model?

- **Highest-quality standard sentence transformer.** `all-mpnet-base-v2` achieves the highest overall score among the standard `all-*` family on the Sentence Embeddings Benchmark and on MTEB clustering tasks. Petukhova et al. (2024) confirm BERT-family transformers are the best open-source option for text clustering.
- **768-dimensional output.** With 110 million parameters and twice the embedding dimensions of MiniLM-L12-v2, the model encodes substantially finer-grained semantic distinctions — essential for separating four linguistically diverse sub-corpora.
- **Cosine-similarity training objective.** The MNR loss trains cosine similarity to be proportional to semantic similarity. This makes the embedding space intrinsically well-suited to cosine-based distance measures, which are used in all subsequent steps.
- **No cloud API dependency.** The model runs fully locally via Hugging Face’s model hub.

2.2 Encoding Configuration

Sentences are encoded in batches of 64 with `normalize_embeddings=False`. Normalisation is applied explicitly in the next step. The output is a `float32` matrix of shape (1760, 768).

Influence on clustering: Sentences from semantically distinct registers — Shakespearean dialogue, contemporary reviews, and encyclopaedic prose — are embedded in clearly separated regions of $\mathbb{R}^{768}$, providing the raw geometric signal from which clusters are recovered.

3. L2 Normalisation

Each raw embedding vector $\mathbf{x} \in \mathbb{R}^{768}$ is L2-normalised:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \quad \text{such that} \quad \|\hat{\mathbf{x}}\|_2 = 1$$

3.1 Why Normalise Before PCA?

Without normalisation, embeddings vary in both direction (semantic content) and magnitude (loosely correlated with sentence length). PCA applied to un-normalised vectors decomposes a mixture of directional and magnitude variance. L2 normalisation ensures PCA decomposes purely **angular (semantic) variance**.

A critical geometric consequence is that, for unit vectors $\hat{\mathbf{a}}, \hat{\mathbf{b}}$:

$$d_{\text{Euclidean}}(\hat{\mathbf{a}}, \hat{\mathbf{b}})^2 = 2(1 - \cos(\hat{\mathbf{a}}, \hat{\mathbf{b}}))$$

Euclidean distance is a **monotone function of cosine distance** on the unit sphere. This ensures that UMAP’s cosine neighbourhood graph and Ward-linkage AHC (Euclidean) are both consistent with the MNR training objective of `all-mpnet-base-v2`.

Influence on clustering: L2 normalisation aligns the distance metric used in every subsequent step (cosine for UMAP neighbourhood graph, Euclidean for PCA projection and AHC) with the metric optimised during model training, ensuring geometric coherence throughout the pipeline.

4. Adaptive Principal Component Analysis

PCA is applied to the L2-normalised matrix $\hat{X} \in \mathbb{R}^{1760 \times 768}$. Rather than fixing a component count, an **adaptive** strategy is used: the minimum number of components n^* whose cumulative explained variance reaches threshold $\tau = 0.80$ is selected:

$$n^* = \min \left\{ n \mid \frac{\sum_{k=1}^n \lambda_k}{\sum_{k=1}^p \lambda_k} \geq \tau \right\}, \quad \tau = 0.80, \quad p = 768$$

For this dataset, $n^* = 121$ components are retained, preserving **80.02%** of total variance.

4.1 Why Apply PCA?

- **Curse of dimensionality.** In $\mathbb{R}^{768}$, pairwise Euclidean distances become nearly uniform (Beyer et al., 1999), degrading distance-based algorithms. Reducing to 121 dimensions substantially mitigates this.
- **Denoising.** Components 122 through 768 capture sentence-level idiosyncrasies rather than

shared semantic structure. Discarding them acts as a low-rank denoising filter.

- **UMAP pre-conditioning.** UMAP’s k -nearest-neighbour graph construction is $O(N \log N)$ in the embedding dimension. Reducing from 768 to 121 dimensions substantially reduces this cost while preserving the semantic structure that UMAP needs to resolve the manifold.

4.2 Why Not Re-Normalise After PCA?

PCA is variance-preserving: PC_1 carries the largest eigenvalue, PC_2 the second largest, and so on. Euclidean distance in PCA space naturally weights the earlier, higher-variance components:

$$d^2(\mathbf{a}, \mathbf{b}) = \sum_{k=1}^{121} (a_k - b_k)^2$$

This variance weighting provides useful inductive bias for UMAP: the most semantically dominant axes (e.g. the contrast between archaic and modern English) automatically receive greater influence in the neighbourhood graph. Re-normalising after PCA would collapse all 121 dimensions to equal contribution, erasing this structure.

Influence on clustering: Adaptive PCA retains exactly the components needed to represent 80% of the semantic variance in the 768-dimensional embedding space, providing UMAP with a clean, noise-reduced, variance-weighted input.

5. UMAP: Non-Linear Manifold Projection

After PCA denoising, Uniform Manifold Approximation and Projection (UMAP; McInnes, Healy & Melville, 2018) projects the 121-dimensional PCA coordinates into a 5-dimensional Euclidean space used directly for clustering.

5.1 Why UMAP After PCA?

PCA is linear: it decomposes global variance along orthogonal axes and cannot represent non-linear cluster structure. Sentence embeddings lie on a curved manifold in $\mathbb{R}^{768}$; sub-domains that are genuinely distinct may not be linearly separable. UMAP constructs a fuzzy topological representation of the data manifold using a k -nearest-neighbour graph, then optimises a low-dimensional layout that preserves both local neighbourhood structure and global topology simultaneously.

5.2 Parameter Choices

- **`metric='cosine'`.** UMAP builds its k -nearest-neighbour graph using cosine distance in the PCA-denoised space. Cosine distance is the semantically correct measure for L2-normalised text embeddings, consistent with the MNR training objective. This is the BERTopic standard (Grootendorst, 2022).
- **`n_components=5`.** Standard output dimensionality for clustering tasks (BERTopic standard). Sufficient to preserve multi-cluster manifold topology; low enough for Euclidean distances to remain well-conditioned for AHC.
- **`min_dist=0.0`.** The UMAP documentation states explicitly that low values are recommended for clustering tasks. `min_dist=0.0` allows points within the same neighbourhood to pack as tightly as the manifold permits, producing compact, isolated cluster regions. This is the BERTopic standard for clustering.

- **n_neighbors=10.** Reduced from the standard default of 15 to increase sensitivity to small, dense local structures. With **n_neighbors=10**, each of the 80 Notre Dame rows finds most of its 10 nearest neighbours within the Notre Dame group itself, creating a strongly connected sub-graph that UMAP projects into a tight, isolated manifold region. The BERTopic documentation explicitly states that decreasing **n_neighbors** “creates more local structure” and is appropriate when detecting small, cohesive sub-domains.
- **random_state=42.** Fixes UMAP’s stochastic gradient descent phase for bit-for-bit reproducibility.

Influence on clustering: UMAP with **n_neighbors=10** is the key step that enables the detection of the Notre Dame sub-domain as a geometrically distinct cluster. The reduced neighbourhood size makes the algorithm more sensitive to the dense, localised cluster formed by the repeated Notre Dame sentences, which UMAP projects into a compact isolated region that Ward-linkage AHC can cleanly separate from the surrounding modern English manifold.

6. Pipeline Summary

Stage	Operation	Input	Output	Key Purpose
1	Data loading	Raw text file	1,760 sentences (dups retained)	Quality & row alignment
2	all-mpnet-base-v2	1,760 sentences	(1760, 768)	Semantic representation
3	L2 normalisation	(1760, 768)	(1760, 768) unit vectors	Cosine-consistent distances
4	Adaptive PCA	(1760, 768)	(1760, 121) variance-weighted	Denoising; UMAP pre-conditioning
5	UMAP	(1760, 121)	(1760, 5) manifold-projected	Local cluster separation

Table 1: Summary of the five-stage preprocessing pipeline.

7. Holistic Discussion: Influence on Clustering Performance

The preprocessing choices form a coherent, theoretically motivated pipeline:

- **Cleaning → Embedding quality.** Preserving natural text keeps sentences within the model’s training distribution. Archaic Shakespeare vocabulary, contemporary review prose, and formal encyclopaedic text produce embeddings in clearly separated regions of $\mathbb{R}^{768}$.
- **Duplicate retention → Notre Dame cluster density.** The 80 Notre Dame rows (18 unique $\times$ 4–5 copies) create a dense local neighbourhood in UMAP space. This is the necessary precondition for AHC to detect the Notre Dame sub-domain as a distinct cluster.

- **L2 normalisation** → **Metric consistency**. All steps from PCA onwards operate in cosine-equivalent Euclidean space, consistent with the MNR training objective.
- **Adaptive PCA** → **UMAP pre-conditioning**. Noise suppression and dimensionality reduction produce a clean, variance-weighted input that allows UMAP to resolve the non-linear manifold structure efficiently.
- **UMAP** ($n_{\text{nbrs}} = 10$) → **Local cluster isolation**. The reduced neighbourhood size amplifies local density differences, enabling the detection of the small but cohesive Notre Dame cluster that would be invisible to a linear projection or to UMAP with larger neighbourhood sizes.

References:

- Beyer, K., Goldstein, J., Ramakrishnan, R., & Shaft, U. (1999). When is “nearest neighbor” meaningful? *ICDT*, 217–235.
- Grootendorst, M. (2022). BERTopic: Neural topic modelling with a class-based TF-IDF procedure. *arXiv:2203.05794*.
- McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv:1802.03426*.
- Petukhova, A., Matos-Carvalho, J.P., & Fachada, N. (2024). Text clustering with LLM embeddings. *arXiv:2403.15112*.
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *Proceedings of EMNLP 2019*, 3982–3992.
- Song, K., Tan, X., Qin, T., Lu, J., & Liu, T.-Y. (2020). MPNet: Masked and permuted pre-training for language understanding. *NeurIPS*, 33, 16857–16867.